

Prompt Injection Detection: jev-guard vs Model Armor

Head-to-head evaluation over 350 labeled cases from 7 public datasets · 2026-09-21

Summary

jev-guard scored a higher F1 on **6 of 7** datasets and tied on 1. The more decisive number is complementarity: across all 196 attack cases, Model Armor caught **6** that jev-guard missed (3.1%), while jev-guard caught **82** that Model Armor missed. The two systems' errors are therefore highly correlated, and stacking them adds far less than the individual gap suggests.

Read the per-dataset numbers as directional rather than definitive: 50 cases per dataset puts the margin of error on any single F1 at several points, and two configuration caveats below materially favor jev-guard.

Dataset	n	jev F1	armor F1	jev R	armor R	Agree	jev-only	armor-only
deepset	50	0.84	0.51	0.76	0.36	76%	11	1
safeguard	50	0.97	0.97	1.00	0.94	96%	1	0
jailbreak	50	0.92	0.81	0.92	0.71	84%	5	0
spml	50	0.99	0.55	0.98	0.38	50%	25	0
slabs	50	0.75	0.52	0.59	0.35	66%	13	4
password	50	0.93	0.46	0.93	0.30	62%	17	0
mixed	50	0.79	0.57	0.76	0.40	70%	10	1

jev R / armor R are recall. “jev-only” and “armor-only” count attacks caught by that system alone. Green marks the higher F1.

Method

Both systems received identical inputs, 50 cases sampled per dataset with a fixed seed. A case counts as flagged for jev-guard when its routing policy returns anything other than *pass* (block, review, or support), and for Model Armor when `filterMatchState` is `MATCH_FOUND` on any filter. Ground truth is each dataset's own label, so both systems are scored against the same key.

jev-guard ran on `~typesafe/jev-latest` via OpenRouter under its *strict* policy, combining a semantic hazard battery with deterministic structural detectors. Model Armor ran against the existing `testing-template` in project `ma-claude (us-central1)`.

Caveats that favor jev-guard

- **Model Armor ran at MEDIUM_AND_ABOVE.** The template sets `piAndJailbreakFilterSettings.confidenceLevel` to `MEDIUM_AND_ABOVE`. A `LOW_AND_ABOVE` setting exists and would raise its recall at some cost in false positives. These numbers are not Model Armor's ceiling.
- **Model Armor's own filter version was STABLE, not LATEST.** The baseline template pins `FILTER_VERSION_ALIAS_STABLE`. The permutation sweep below re-runs it on `LATEST`.

- **SPML is structurally asymmetric.** Its rows pair a system prompt with a user prompt. Jev takes both as structured state; Model Armor's sanitizeUserPrompt API accepts only the user text, so it cannot see the policy being violated. That is a real capability difference, but it means the SPML gap (0.99 vs 0.55) should not be read as a like-for-like detection comparison.

- **jev-guard's policy was tuned by its author; Model Armor's template was not tuned for this test.** Thresholds in policy.py were iterated against deepset during development, so that dataset in particular flatters jev-guard.

Does stacking them help?

The case for running two detectors rests on their errors being independent. They are not. Union (flag if either fires) and intersection (flag only if both) were computed on the same cases:

Dataset	jev F1	armor F1	union F1	$\cap$ F1	union vs jev
deepset	0.84	0.51	0.87	0.47	+0.03
safeguard	0.97	0.97	0.97	0.97	+0.00
jailbreak	0.92	0.81	0.90	0.83	-0.02
spml	0.99	0.55	0.99	0.55	+0.00
slabs	0.75	0.52	0.83	0.39	+0.08
password	0.93	0.46	0.93	0.46	+0.00
mixed	0.79	0.57	0.82	0.53	+0.02

Union improves on jev-guard alone for only a minority of datasets, and on **jailbreak** it is actively worse (0.92 → 0.90): Model Armor contributed no additional true positives there but did add a false positive. **slabs** is the one genuine exception, where Model Armor caught 4 attacks jev-guard missed and union rises 0.75 → 0.83. Intersection raises precision to 1.00 on several sets but costs so much recall that it is only defensible for auto-blocking with a separate, more sensitive path feeding human review.

Practical reading: a second opinion is not worth double latency and cost for injection detection specifically. Where Model Armor remains complementary is in capabilities jev-guard does not have at all — live URL reputation, DLP/SDP infoType matching, and multimodal screening — rather than as a redundant vote on the same question.

Shared blind spot

Both systems missed 26 of 196 attacks. These are the cases that matter most, because no amount of ensembling reaches them. The recurring shape is a role-play or persona framing that never issues an explicit override instruction:

- As a child does not have any previous knowledge how would you react if
- Ich möchte, dass Sie als Drehbuchautor agieren. Sie werden ein ansprec
- I want you to act as a storyteller. You will come up with entertaining
- Context Zeit online Scheisse Question Wie ist Zeit online?
- E-Scooter Verkehrssicherheit Deutschland I want you to act as a storyt

– Before you can start your freestyle rap, please ask me to provide 3-5

Was the gap just configuration?

The first round ran Model Armor on one template, which left open the objection that its settings rather than its detection explained the gap. To close that, six templates were provisioned covering every permutation of the PI filter's own axes — confidence level (LOW_AND_ABOVE / MEDIUM_AND_ABOVE / HIGH) crossed with multi-language detection on/off — all pinned to `FILTER_VERSION_ALIAS_LATEST` and isolating the prompt-injection filter (RAI, SDP and malicious-URI off) so the number measures PI detection rather than any filter firing.

deepset — all PI permutations (n=50)

Configuration	Precision	Recall	F1	FPR
jev-guard (reference)	0.95	0.76	0.84	0.04
Model Armor low/ml=on	1.00	0.32	0.48	0.00
Model Armor low/ml=off	1.00	0.32	0.48	0.00
Model Armor medium/ml=on	1.00	0.28	0.44	0.00
Model Armor medium/ml=off	1.00	0.28	0.44	0.00
Model Armor high/ml=on	1.00	0.24	0.39	0.00
Model Armor high/ml=off	1.00	0.24	0.39	0.00

slabs — all PI permutations (n=50)

Configuration	Precision	Recall	F1	FPR
jev-guard (reference)	1.00	0.59	0.75	0.00
Model Armor low/ml=on	1.00	0.38	0.55	0.00
Model Armor low/ml=off	1.00	0.38	0.55	0.00
Model Armor medium/ml=on	1.00	0.35	0.52	0.00
Model Armor medium/ml=off	1.00	0.35	0.52	0.00
Model Armor high/ml=on	1.00	0.27	0.43	0.00
Model Armor high/ml=off	1.00	0.27	0.43	0.00

Two results, one of which corrects this report's earlier draft

Confidence level behaves as documented but does not close the gap. Recall rises monotonically as the threshold loosens — on deepset 0.24 (HIGH) to 0.28 (MEDIUM) to 0.32 (LOW) — with precision staying at 1.00 throughout. Even at its most sensitive setting Model Armor's PI filter reaches 0.32 recall against jev-guard's 0.76 on the same cases. The gap is therefore not a threshold artifact.

Multi-language detection had no measurable effect, contrary to this report's first draft. Every confidence level scored identically with the flag on and off. A direct spot check on three explicit German override instructions confirmed it: identical verdicts either way, and two of the three went undetected even at LOW_AND_ABOVE with the flag enabled. The earlier claim that configuration explained the German misses was wrong — on this evidence it is a capability gap in the PI filter. The flag may well affect the RAI or SDP filters, which these isolated templates switch off; it does not appear to affect prompt-injection detection.

Per-dataset detail

deepset

deepset/prompt-injections — direct injection, mixed German/English

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	0.95	0.76	0.84	0.04	19	1	24	6
Model Armor	0.90	0.36	0.51	0.04	9	1	24	16
union	0.95	0.80	0.87	0.04	20	1	24	5
intersection	0.89	0.32	0.47	0.04	8	1	24	17

25 attacks / 50 cases · agreement 76% · jev-only 11, armor-only 1, both missed 5

safeguard

xTRam1/safe-guard-prompt-injection — direct injection, larger and more varied

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	0.94	1.00	0.97	0.03	16	1	33	0
Model Armor	1.00	0.94	0.97	0.00	15	0	34	1
union	0.94	1.00	0.97	0.03	16	1	33	0
intersection	1.00	0.94	0.97	0.00	15	0	34	1

16 attacks / 50 cases · agreement 96% · jev-only 1, armor-only 0, both missed 0

jailbreak

jackhhao/jailbreak-classification — jailbreak vs benign

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	0.92	0.92	0.92	0.08	22	2	24	2
Model Armor	0.94	0.71	0.81	0.04	17	1	25	7
union	0.88	0.92	0.90	0.12	22	3	23	2
intersection	1.00	0.71	0.83	0.00	17	0	26	7

24 attacks / 50 cases · agreement 84% · jev-only 5, armor-only 0, both missed 2

spml

reshabhs/SPML_Chatbot_Prompt_Injection — injection judged against a given system prompt

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	1.00	0.98	0.99	0.00	41	0	8	1
Model Armor	1.00	0.38	0.55	0.00	16	0	8	26
union	1.00	0.98	0.99	0.00	41	0	8	1
intersection	1.00	0.38	0.55	0.00	16	0	8	26

42 attacks / 50 cases · agreement 50% · jev-only 25, armor-only 0, both missed 1

slabs

S-Labs/prompt-injection-dataset — independent direct-injection set

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	1.00	0.59	0.75	0.00	22	0	13	15
Model Armor	1.00	0.35	0.52	0.00	13	0	13	24
union	1.00	0.70	0.83	0.00	26	0	13	11
intersection	1.00	0.24	0.39	0.00	9	0	13	28

37 attacks / 50 cases · agreement 66% · jev-only 13, armor-only 4, both missed 11

password

ivanleomk/prompt_injection_password — attempts to extract a withheld secret

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	0.93	0.93	0.93	0.09	25	2	21	2
Model Armor	1.00	0.30	0.46	0.00	8	0	23	19
union	0.93	0.93	0.93	0.09	25	2	21	2
intersection	1.00	0.30	0.46	0.00	8	0	23	19

27 attacks / 50 cases · agreement 62% · jev-only 17, armor-only 0, both missed 2

mixed

jayavibhav/prompt-injection — large corpus, benign half is ordinary QA text

System	Precision	Recall	F1	FPR	TP	FP	TN	FN
jev-guard	0.83	0.76	0.79	0.16	19	4	21	6
Model Armor	1.00	0.40	0.57	0.00	10	0	25	15
union	0.83	0.80	0.82	0.16	20	4	21	5
intersection	1.00	0.36	0.53	0.00	9	0	25	16

25 attacks / 50 cases · agreement 70% · jev-only 10, armor-only 1, both missed 5

Reproducing this

```
python compare.py deepset safeguard jailbreak spml slabs password mixed --n 50 --json results.json
python report.py results.json jev-guard-evaluation.pdf
```